

ME594: 3D Robotics, HW 9

Homework due on 03-23-2025, Topics: Trajectory Generation. Please read each question on how to submit the work.

1. **Trajectory with velocity constraint:** Consider a scenario where the following are specified: initial position is q_0 at time $t = 0$, the final position is q_f at time $t = t_f$, and the maximum speed V . We want a velocity profile shown in Fig. 1 (b): a ramp up from 0 to t_a , a constant speed of V from t_a to $t_f - t_a$, and a ramp down from $t_f - t_a$ to t_f . Compute

- (a) Compute $q(t)$ from 0 to t_f (Fig. 1 (a)).
(b) What is the condition on V such that the velocity will reach a steady state of V ?

HINT:

- (i) You need to choose three-time profiles: first from 0 to t_a , $\dot{q}(t) = c_0 + c_1t$, second from t_a to $t_f - t_a$, $\dot{q}(t) = V$, and third from $t_f - t_a$ to t_f , $\dot{q} = c_2 + c_3t$.
(ii) You may use t_a and a (=acceleration, see Fig. 1 (c)) to help you do the calculations, but you need to compute them as functions of known quantities (q_0 , q_f , t_f , and V)
(iii) The symmetry of the $\dot{q}$ vs t will help to compute t_a and a . (e.g., $q(t = 0.5t_f) = 0.5(q_0 + q_f)$).

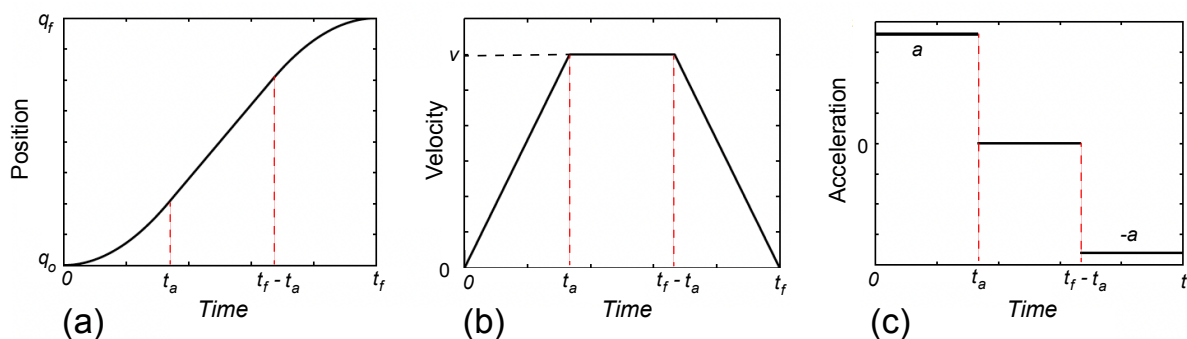


Figure 1: Trajectory generation for constant velocity profile.

- **Gradescope:** Submit your answer to both parts and any related work. Hand calculations and/or symbolic computations are both acceptable.
- **Email:** None

2. **Trajectory generation for Trossen Robotics WidowX 250 S:** Figure 2 shows the WidowX 250S by Trossen Robotics. A video of the robot in action is here: https://www.youtube.com/watch?v=l_10IH0JNxI. I have provided the *wx250s_mocap.xml* file. Please use this file and include it in the scene.xml meshes that can be downloaded from GitHub

https://github.com/google-deepmind/mujoco_menagerie/tree/main/trossen_wx250s. I have also provided you with a folder *wx250s_traj-generation* that contains the main file *mj_wx250s_traj-gen-cart.py* which will run fine once you set the correct *xml path* (top of the file). I have put HINTS in this file which you will help you solve this problem.

You aim to get the yellow block (this not a real box but a mocap shape for tracking the end-effector position, see line 111 in the xml file) from the initial position $(0.4, 0.25, 0.02)$ onto the top of the green box without colliding with the obstacles (green box, brown wall), the floor, or the manipulator. You will follow a two part strategy.

- (a) First, plan the cartesian motion of the yellow box. This can be done by setting its position using *data.mocap_pos* and its orientation using *data.mocap_quat*. Use a suitable interpolation for the position and rotation. See HINT1 and HINT2 in *mj_wx250s_traj-gen-cart.py*
- (b) Second, plan the joint angle motion using the cartesian motion of the yellow box as a reference to the inverse kinematics. See HINT3 in *mj_wx250s_traj-gen-cart.py*

HINTS

- i You need to figure out the end position of the box so that it is placed on the green box. The center of the top face of the green box is $(0.4, -0.2, 0.3)$ and the dimensions of the top face are 0.2×0.1 .
- ii You need to figure out the start and end orientation of the yellow block.
- iii You might have to plan intermediate points if a single trajectory leads to a collision.
- iv Here is my attempt <https://youtu.be/hqdxoyWRGn8>.

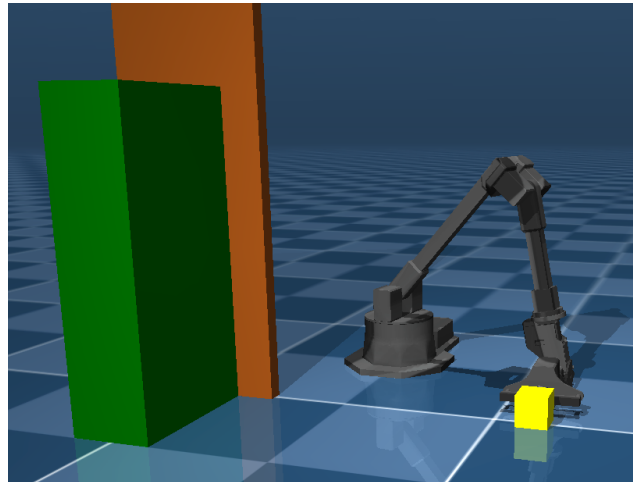


Figure 2: Trajectory generation for the Trossen WidowX 250 S

Submission:

- **Gradescope:** None.
- **Email:** Submit a zipped folder: *w250s_traj_control_names.zip* where *names* should be replaced by names of the team member(s) and sent to pranav@uic.edu. Please add “HW10 wx250s” to your subject line. If submitting as a group, please copy your group member.